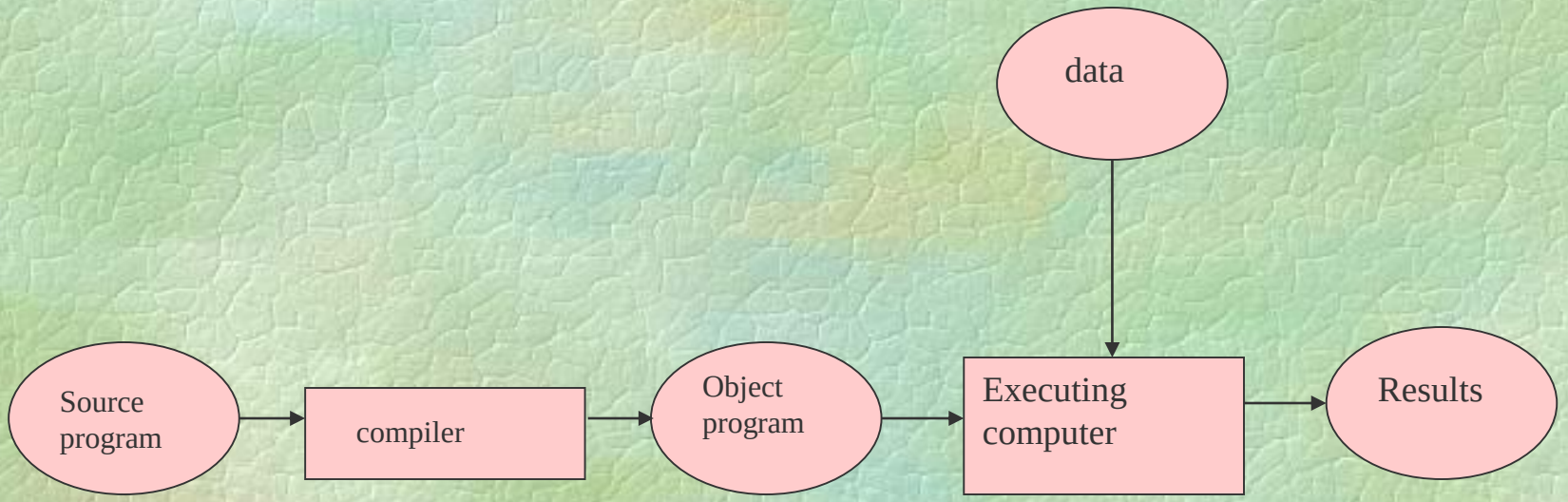
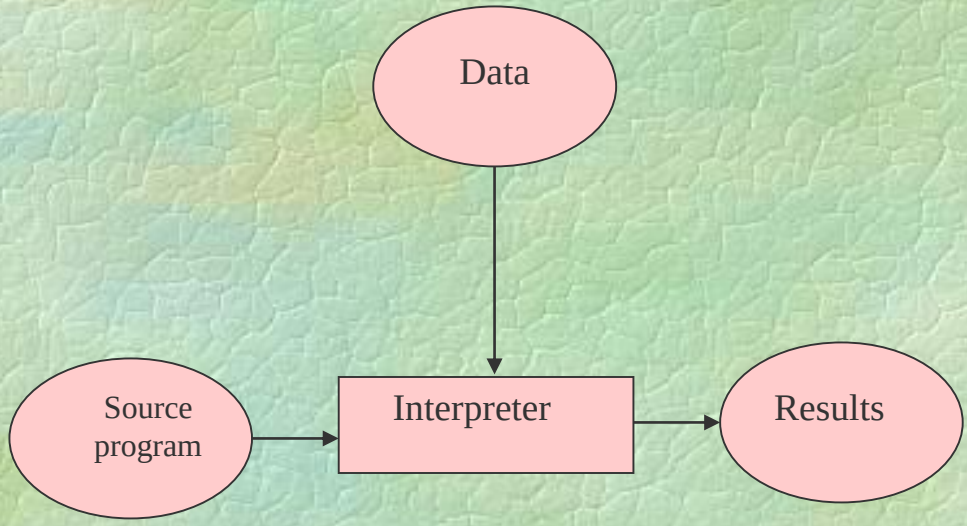
Language Processors

- In order to run /to execute/, any program written in HLL (incl. Java) should be transformed to executable form
- Three ways to convert source code into executable form:
  - Compiler – generate object code – see slide+1
  - Interpreter – interpret source code – see+2
  - Compiler of hybrid (semi-interpreting) type - +3



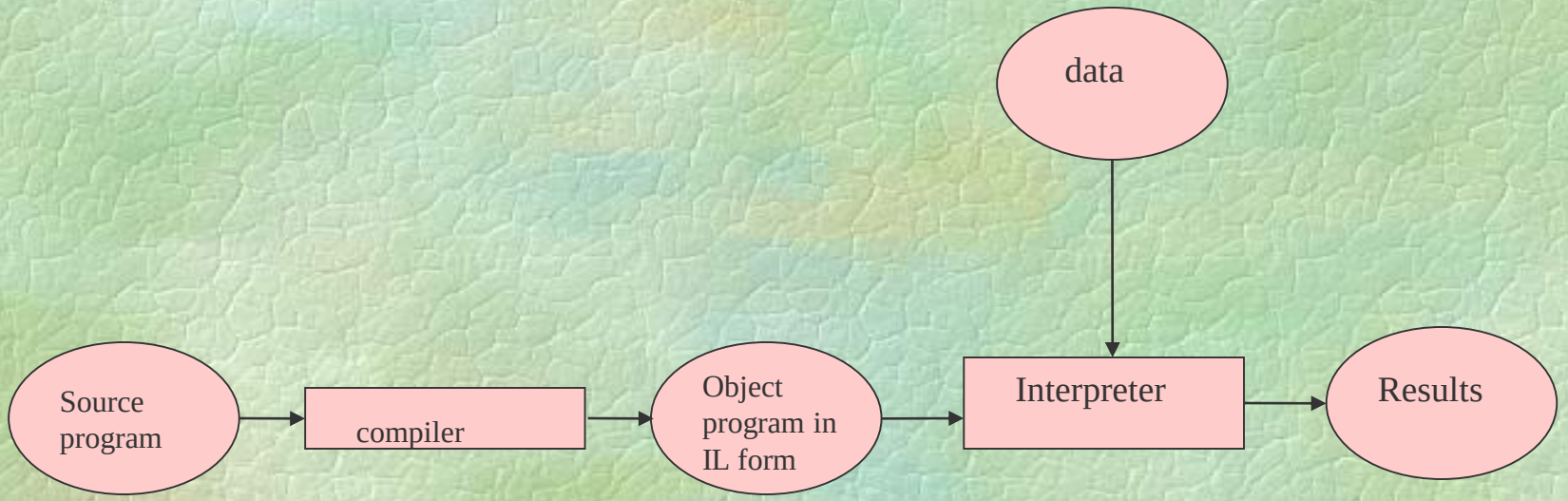
Compile time

Run time



Compile time

Run time



Compile1 time

Compile2 time

Run time

How Java works

Java compiled to intermediate form – byte code.

Byte code further processed:

by interpreter named JVM

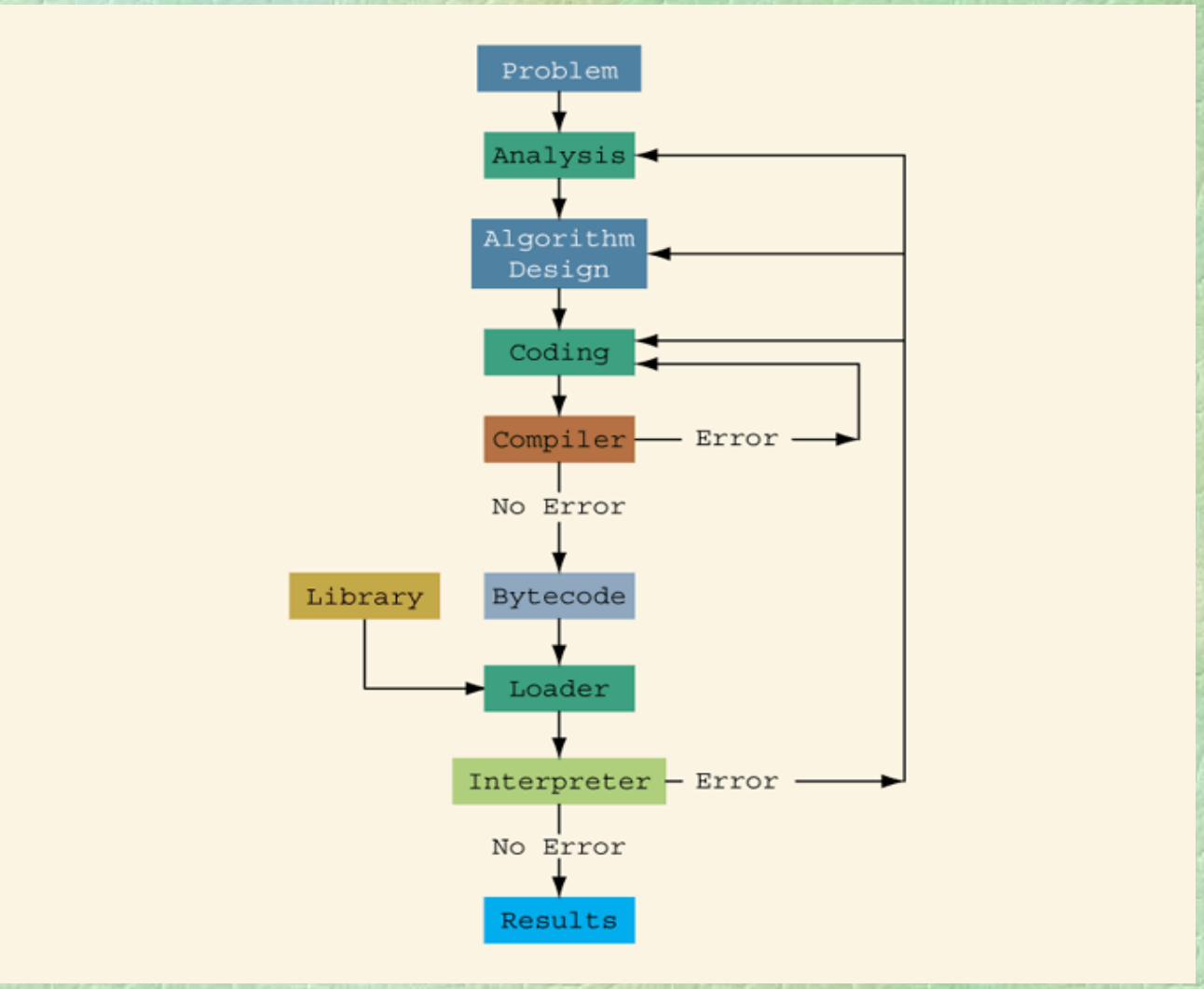
OR

by JIT compiler.

Reminder and Refresh

The SDLC concept – see next slide

Problem-Analysis-Coding-Execution Cycle



Java Components of special interest

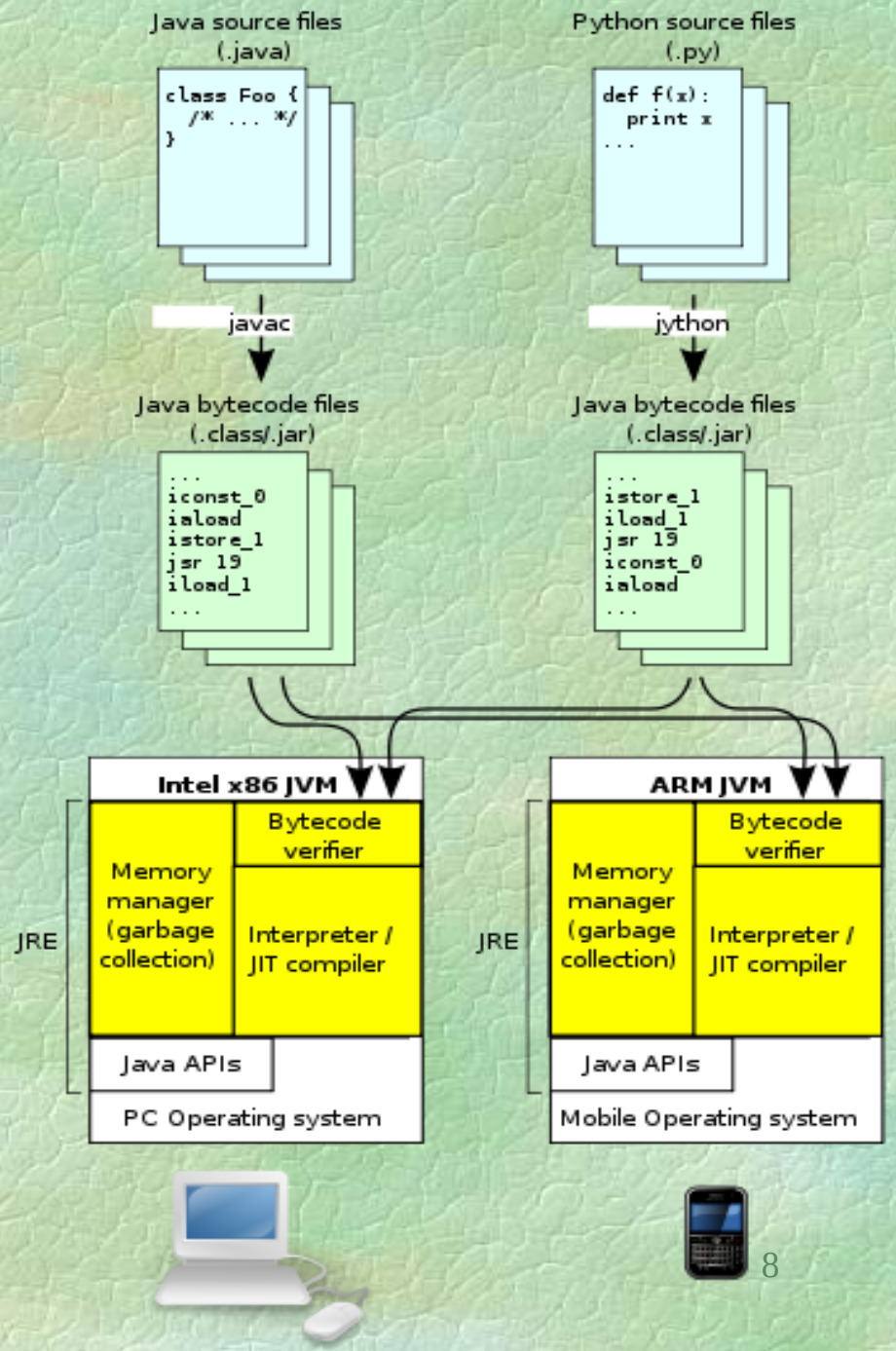
- Pure Java includes 3 software facilities:
- JRE (includes JVM)
- JVM (a part of JRE)
- JDK

JVM

- JVM architecture.

  Source code is compiled to Java bytecode, which is verified, interpreted or JIT-compiled for the native architecture.

  The Java APIs and JVM together make up the Java Runtime Environment (JRE).



JDK contents

The JDK has
as
its primary components
a collection
of programming tools,
including:

JDK contents (most often used utilities)

- appletviewer – this tool can be used to run and debug Java applets without a web browser
- java – the loader for Java applications. This tool is an interpreter and can interpret the class files generated by the javac compiler.
- javac – the Java compiler, which converts source code into Java bytecode
- javadoc – the documentation generator, which automatically generates documentation from source code comments
- jar – the archiver, which packages related class libraries into a single JAR file. This tool also helps manage JAR files.

A Simple Java Program

Listing 1.1

```
//This program prints message “Welcome to Java!“
// braces in end-line style

public class Welcome {
  public static void main(String[] args) {
    System.out.println("Welcome to Java!");
  }
}
```

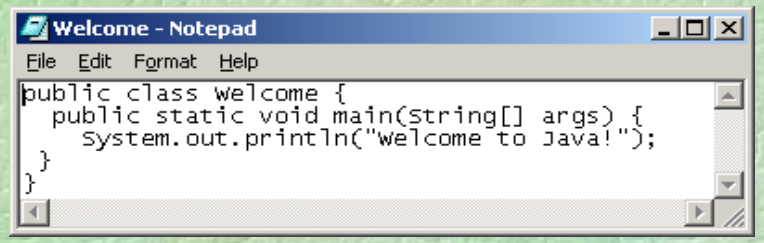
A Simple Java Program

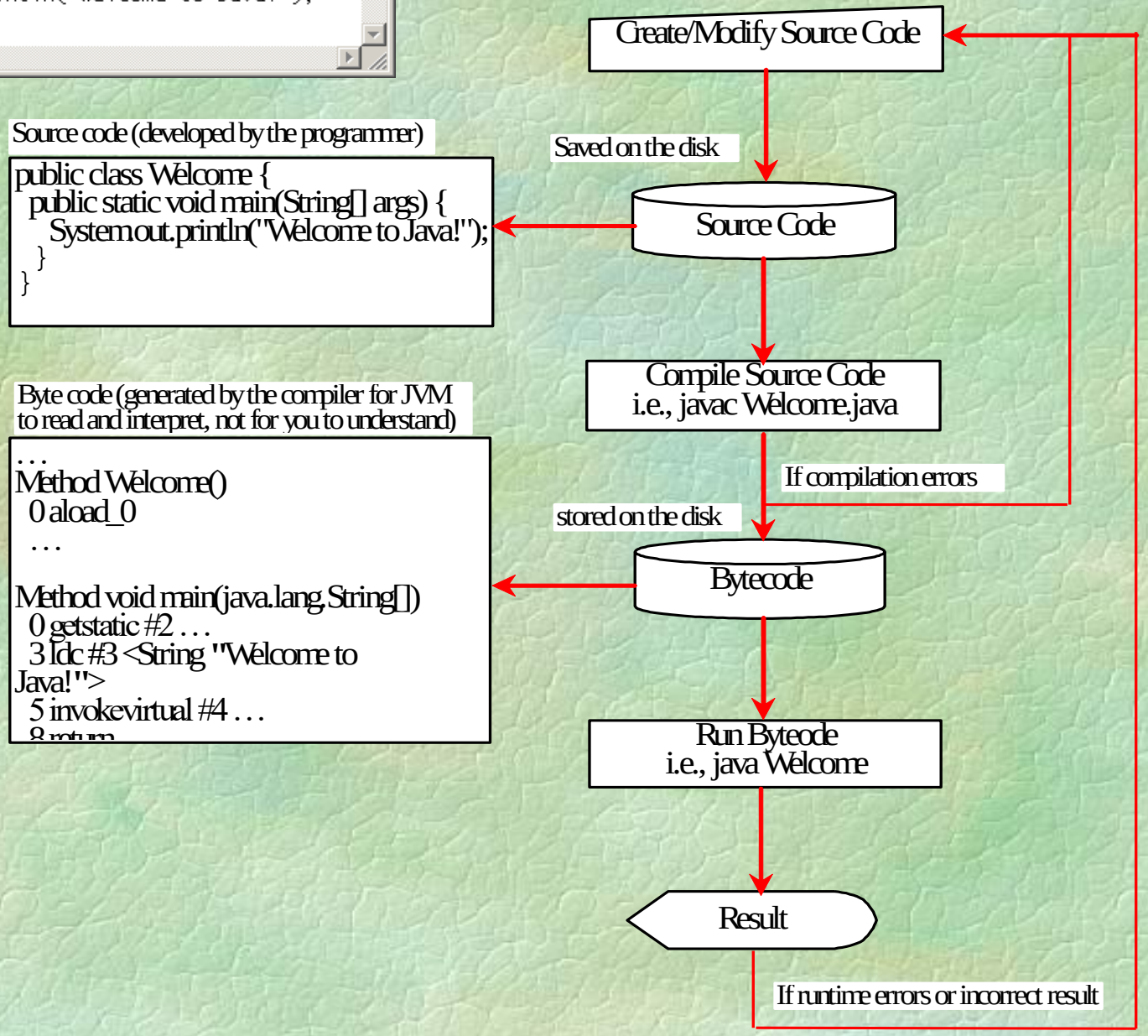
Listing 1.1

```
//This program prints message “Welcome to Java!“
// braces in new-line style

public class Welcome
{
  public static void main(String[] args)
  {
    System.out.println("Welcome to Java!");
  }
}
```

Creating, Compiling, and Running Java Programs





Running JAVA in Windows Environment

Creating, Compiling & Running Java Programs

- From the Command Window
  - Details follow

To open command window

- Click Start
- Select All Programs >
- Select Accessories
- Click Command Prompt

Compiling and Running Java programs

- Using any text editor , /notepad, edit, write/ type the source text of a Java demo program like this:

```
public class prog4 {
 public static void main(String[] args){
   System.out.println("Hello from Java");
 }
}
```

- Save the Java program as a prog4.java file.

Compiling and Running Java programs

- Run compiler with statement like this:

```
javac prog4.java
```

- Run application with stmt like this:

```
java prog4
```